

Subject Description Form

Subject Code	EEE513
Subject Title	Introduction to Quantum Computing
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	The objective of the subject "Introduction to Quantum Computing" is to provide students with a foundational understanding of the principles, concepts, and applications of quantum computing. To develop a strong foundation in quantum mechanics, quantum algorithms, and quantum programming, enabling them to understand the potential and limitations of quantum computing technology.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Demonstrate a foundational understanding of key principles in quantum mechanics, including basic quantum gates and circuits to perform quantum computations. - Describe and explain essential quantum algorithms such as Shor's algorithm and Grover's algorithm. - Identify different quantum computing hardware platforms and their underlying technologies. - Recognize the importance of quantum error correction in maintaining the integrity of quantum information. - Utilize quantum programming languages and tools to write and execute quantum programs. - Evaluate potential applications of quantum computing in cryptography, optimization, and other domains.
Subject Synopsis/ Indicative Syllabus	Basics of Quantum Mechanics 1-1 Overview of classical vs. quantum mechanics 1-2 Quantum states, qubits, and quantum gates 1-3 Superposition and measurement Quantum Gates and Circuits 2-1 Quantum gates: Pauli-X, Pauli-Y, Pauli-Z, Hadamard 2-2 Building quantum circuits and circuit composition 2-3 Unitary transformations and quantum state evolution Quantum Algorithms 3-1 Quantum parallelism and quantum speedup 3-2 Shor's algorithm for factoring large numbers 3-3 Grover's algorithm for unsorted database search

	4. Quantum Hardware and Technologies 4-1 Types of quantum hardware: superconducting qubits, trapped ions, etc. 4-2 Quantum coherence, decoherence, and error rates 4-3 Quantum gates and gate fidelity 5. Quantum Error Correction 5-1 Introduction to quantum error correction 5-2 Quantum error models and error channels 5-3 Basics of stabilizer codes and fault tolerance 6. Quantum Programming Languages 6-1 Introduction to quantum programming languages: Qiskit, Cirq 6-2 Writing and simulating simple quantum programs 6-3 Quantum state visualization and quantum circuits 7. Applications of Quantum Computing 7-1 Quantum cryptography and secure communication 7-2 Quantum optimization and search problems 7-3 Quantum simulations in material science and chemistry 8. Challenges and Future Directions 8.1 Quantum supremacy and benchmarks 8.2 Scalability challenges and quantum hardware limitations 8.3 Emerging trends in quantum computing research																																						
Teaching/Learning Methodology	Lecture: The fundamentals of quantum computing algorithms and hardwares will be described. Students are free to request help. Students are encouraged to solve problems and to use their knowledge to verify their solutions before seeking assistance. Tutorial: A set of problems and group discussion topics will be arranged in the tutorial classes. Students are encouraged to solve problems before having solutions.																																						
Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed (Please tick as appropriate)</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Continuous Assessments (Assignments and tests)</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>2. Examination</td><td>60%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Continuous assessments consist of assignments and mid-term test. These will assess the students’ understanding of basic concepts and principles in quantum communication technologies. The examination will be conducted to make a comprehensive assessment of students’ intended learning outcomes as stated above.	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)						a	b	c	d	e	f	1. Continuous Assessments (Assignments and tests)	40%	✓	✓	✓	✓	✓	✓	2. Examination	60%	✓	✓	✓	✓	✓	✓	Total	100 %						
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2. Examination	60%	✓	✓	✓	✓	✓	✓																																
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Student Study Effort Expected	Class contact:	
	▪ Lecture	33 Hrs.
	▪ Tutorial	6 Hrs
	Other student study efforts:	
	▪ Reading and self-study	81 Hrs.
	The total student study effort	120 Hrs.
Reading List and References	Text Books • Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010. • Rieffel, Eleanor G., and Wolfgang H. Polak. Quantum computing: A gentle introduction. MIT Press, 2011. Reference Book • Johnston, Eric R., Nic Harrigan, and Mercedes Gimeno-Segovia. Programming quantum computers: essential algorithms and code samples. O'Reilly Media, 2019.	